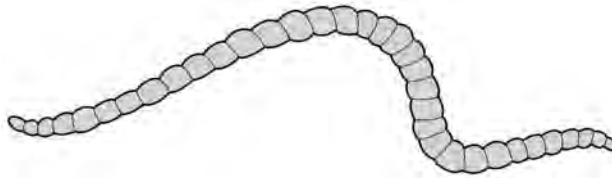


**0 3**

Tubifex worms are small, thin animals that live in water. They have no specialised gas exchange or circulatory system.

**Figure 2** shows a tubifex worm.

**Figure 2**

**0 3 . 1**

Name the process by which oxygen reaches the cells inside the body of a tubifex worm.

**[1 mark]**

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**0 3 . 2**

Using the information provided, explain how **two** features of the body of the tubifex worm allow efficient gas exchange.

**[2 marks]**

1 

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2 

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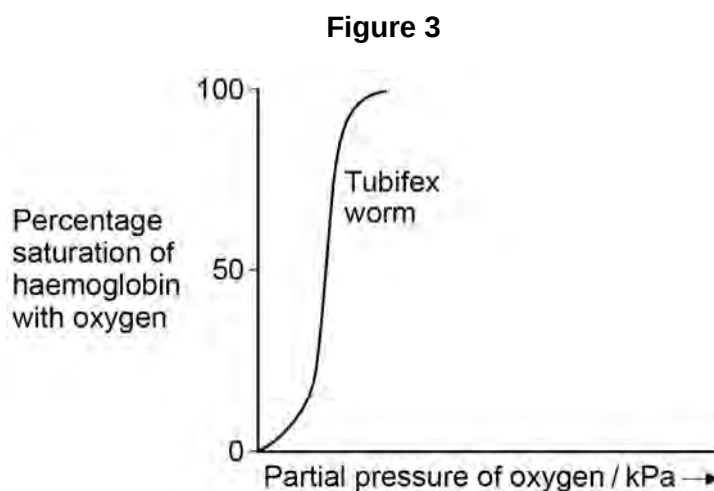
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0 3 . 3

Most species of tubifex worms live at the bottom of ponds, lakes and rivers where the partial pressure of oxygen is low. Pollution of water by sewage can cause the partial pressure of oxygen to fall below 0.2 kPa.

**Figure 3** shows the oxyhaemoglobin dissociation curve for a species of tubifex worm found in a river polluted with sewage.



The species of tubifex worm in **Figure 3** has 50% saturation of their haemoglobin with oxygen at 0.08 kPa.

Explain how this enables this species to survive in water polluted with sewage.

[2 marks]

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0 3 . 4

Species of tubifex worm that live in ponds, lakes and rivers cannot survive in seawater.

Use your knowledge of water potential to explain why they **cannot** survive in seawater.

[2 marks]

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